

SANTIAGO NEIRA LOPEZ

University of Massachusetts Amherst
Department of Mathematics, LGRT 1235M
+1 413 4661923

Website — sneiralopez@umass.edu — linkedin.com/in/thiagoneira/ — github.com/ThiagoNeira

PROFESSIONAL SUMMARY

PhD student in Mathematics specializing in statistics, machine learning, and scientific computing. Experienced developing probabilistic models, deep learning methods, and numerical algorithms for large-scale scientific applications. Interested in Machine Learning, Data Science, and Quantitative Research internships.

TECHNICAL SKILLS

Programming: Python, R

Libraries: PyTorch, NumPy, SciPy, Pandas, GPy, Emukit

Machine Learning: Linear Models, Gaussian Processes, Bayesian Optimization, Deep Learning, Operator Learning, Generative Models, Probabilistic Machine Learning

Statistics: Statistical Inference, Bayesian Statistics, Probability Theory, Stochastic Processes

Applied Mathematics: Applied Mathematics: Numerical Methods, Optimization, PDEs.

EXPERIENCE

Graduate Teaching Assistant

University of Massachusetts Amherst, 2023-Present.

- Led weekly discussion sections for 30–40 undergraduate students in Calculus I and II.
- Designed problem-solving sessions and provided one-on-one support, improving students' understanding of core mathematical concepts.
- Communicated complex mathematical ideas to students with diverse backgrounds.

PROJECTS

Heston Stochastic Volatility Model (Collaborative Project)

- Implemented European option pricing algorithms under the Heston stochastic volatility model in Python.
- Developed a calibration pipeline combining global and local optimization methods.
- Applied the calibration framework to real TSLA options market data.
- Benchmarked pricing accuracy and computational efficiency across numerical methods.
- Technologies: Python, NumPy, SciPy, Pandas
- Repository: [GitHub](#)

EDUCATION

PhD in Mathematics

University of Massachusetts Amherst, 2023–Present.

- Research Interest: Machine Learning, Statistics, Surrogate Modeling

MSc in Mathematics

University of Massachusetts Amherst, MA USA, 2023–2025.

BSc in Mathematics

Pontificia Universidad Javeriana, Bogota Colombia, 2019–2023.

CERTIFICATIONS

Summer 2026 Quant Finance Boot Camp — The Erdős Institute Jun 2026

SELECTED COURSES

- Machine Learning
- Bayesian Statistics
- Statistical Inference
- Probability
- Stochastic Processes
- Numerical methods
- Mathematics of Generative AI

HONORS & AWARDS

- 2023. Orden al Mérito Académico Javeriano
- 2023. Best GPA, Honors for Bachelor's Thesis